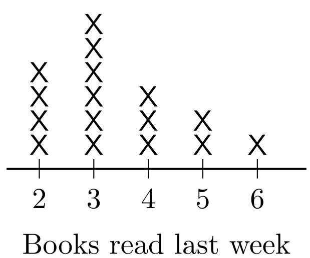
Reading and Making Line Plots

Reading X's off a whole-number line plot, drawing the missing X's to finish one, and comparing or totalling what the X's show

How to use this sheet.

- A *line plot* is a number line with one X for every thing that was counted or measured. The numbers go along the bottom in order, evenly spaced.
- One X always stands for one thing. To find out how many things had a certain number, do not read the number — count the X's stacked above it.
- A number with no X's above it still belongs on the line. It tells you that nothing landed there.
- Touch each X with your pencil as you count it, and say the count out loud. Counting a stack twice is the mistake this sheet keeps coming back to.

1. **Count one stack.** Room 4 made a line plot of how many books each child read last week.

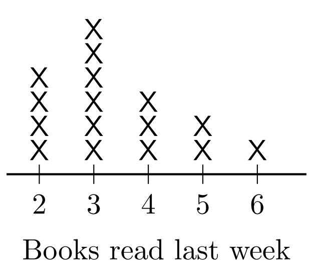


Each X is one child.

Put your pencil on the 4 at the bottom. Now count straight up.

How many children read exactly 4 books? _________

2. **Find the tallest stack.** Same line plot, same room.

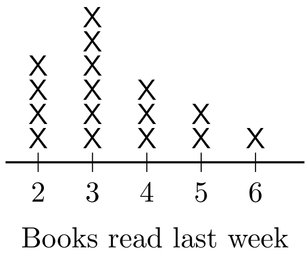


The tallest stack shows the number of books that the *most* children read.

Which number of books has the tallest stack? _________

How many X's are in that stack? ______

3. How many children in all? Every child in Room 4 put exactly one X on the plot.



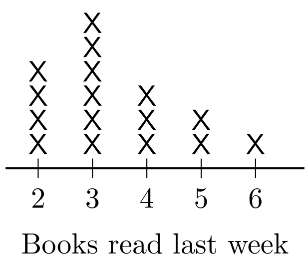
Write the size of each stack first, then add them.

Above 2: ________ Above 3: ________
Above 4: ________

Above 5: ________ Above 6: ________

Children in Room 4 altogether: ________

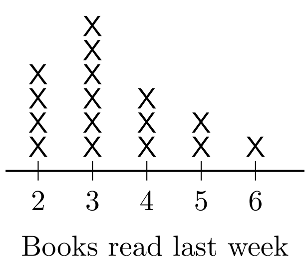
4. Find the shortest stack. Careful — this is not the number furthest to the left.



Which number of books was read by the *fewest* children? ________

How many children was that? ______

5. How many more? Compare two stacks without counting the whole plot.



"How many more" means you take one stack away from the other.

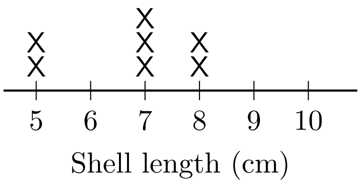
X's above 3: ________

X's above 4: ________

How many more children read 3 books than 4 books? ________

6. Finish the line plot. Ms. Ruiz measured a tray of shells to the nearest whole centimetre. Here are the lengths she wrote down:

5, 7, 6, 8, 6, 5, 7, 6, 10, 7, 6, 8

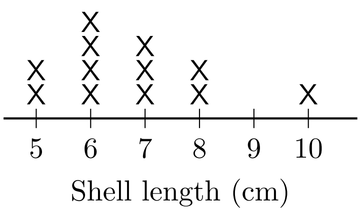


The X's for the 5's, the 7's and the 8's are already drawn. Cross each of those off the list, then draw the X's that are still missing.

How many X's did you draw above 6? _________

Above 10? ______

7. How many shells? Here is the finished plot. Check yours against it before you go on.

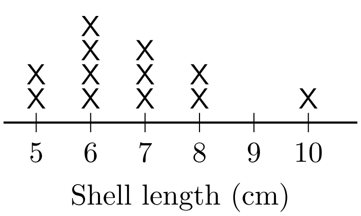


Do not count the numbers along the bottom. Count the X's.

How many shells did Ms. Ruiz measure in all? _________

Does your answer match how many lengths were on her list? ________

8. Compare two stacks. Same finished plot.

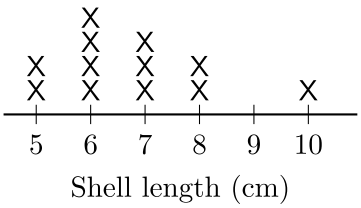


The question asks about *shells*, so both numbers you need are stack sizes.

X's above 6: _________ X's above 5: _________

How many more shells were 6 cm long than 5 cm long? _________

9. Longest and shortest. Same finished plot. This time the answer is a length, so it is measured in centimetres.



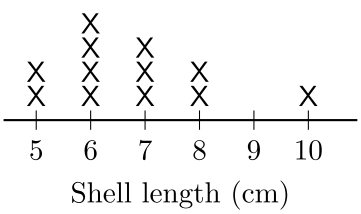
Find the X that is furthest right and the X that is furthest left.

Longest shell: ______ Shortest shell: ______

How much longer was the longest shell than the shortest one?

Answer: ________________ cm

10. Explain it to Sam. Sam looks at the finished plot and says: "There are no X's above the 9, so Ms. Ruiz forgot that one. We should rub the 9 off the line."



(a) How many shells were 9 cm long? _________

(b) Write one or two sentences telling Sam what the empty space above the 9 really means, and why the 9 has to stay on the line.

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